

MODULE 7A • EPISODE 2

Our Cosmic Neighborhood: Planets, Moons, and Galactic Perspectives

Mars, Venus, the Moon, and Our Place Among the Galaxies

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Our Cosmic Neighborhood: Planets, Moons, and Galactic Perspectives

The previous episode stayed close to home with timekeeping, the seasons, and the autumn and winter sky. This one widens the view in steps. It begins with our two planetary neighbors — Mars, cold and thin-aired, and Venus, crushing and scorching — then turns to our own Moon: how it formed, why it keeps drifting away, why we only ever see one side of it, and how its phases follow a predictable schedule. From there it steps out to the Milky Way, the Local Group of galaxies, and the giant elliptical galaxy M87, whose central black hole was the first ever imaged. It closes back on Earth, with the sailors who found their latitude from the North Star.

The Panel

Speaker	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Cosmologist, European Space Agency
Dr. Amara Johnson	Theoretical physicist

Learning Objectives

After this episode you should be able to:

- ◆ Describe Mars's color, moons, magnetic environment, day, year, seasons, atmosphere, and Olympus Mons.
- ◆ Explain why Venus is seen only as a morning or evening star, and recall its ancient names.
- ◆ Explain why Venus's surface is so hostile, and why its day is about half its year.
- ◆ Account for the 16% weight reduction a visitor would feel on Venus.
- ◆ Describe the giant-impact origin of the Moon and the evidence for it.
- ◆ Explain why the Moon is moving away from Earth, and how that distance is measured.
- ◆ Distinguish a **sidereal month** from a **lunation**, and explain tidal locking.
- ◆ Predict when and where the main lunar phases appear in the sky.
- ◆ Describe the Milky Way, the Local Group, and elliptical galaxies such as M87.
- ◆ Explain how the altitude of Polaris gives an observer's latitude.

QUICK REFERENCE

Key Terms

Term	Definition
Conjunction	An alignment in which a planet lies in nearly the same direction as the Sun and is lost in its glare.
Iron oxide	Rust. It coats the surface of Mars and gives the planet its red color.
Magnetosphere	The region around a planet controlled by its magnetic field, which deflects the solar wind.
Solar wind	The stream of charged particles flowing outward from the Sun.
Sol	A Martian day, just under 25 Earth hours long.
Terrestrial planet	A rocky planet: Mercury, Venus, Earth, or Mars.
Astronomical unit (AU)	The average Earth-Sun distance, about 93 million miles.
Inferior planet	A planet closer to the Sun than Earth is; seen only in the morning or evening sky.
Runaway greenhouse effect	Heat trapping that feeds on itself until a planet's surface is extremely hot, as on Venus.
Bar	A unit of pressure equal to 100,000 pascals, close to Earth's sea-level air pressure.
Lux	A unit measuring how brightly a surface is illuminated.
Retrograde rotation	Spinning in the direction opposite to that of Earth and most other planets.
Buoyant force	The upward push a fluid, including air, exerts on an object immersed in it.
Theia	The Mars-sized body whose glancing collision with the early Earth produced the Moon.
Tidal locking	The state in which a body's rotation period equals its orbital period, so it always shows the same face to its partner.
Sidereal month	The time for the Moon to orbit Earth once relative to the distant stars: 27.3 days.
Lunation	The time from one New Moon to the next, also called a synodic month: 29.5 days.
Terminator	The line between the sunlit and dark parts of the Moon (or any world).
Barred spiral galaxy	A spiral galaxy whose arms begin at the ends of a bar of stars through its center. The Milky Way is one.
Local Group	The gravitationally bound family of galaxies that includes the Milky Way and Andromeda.
Elliptical galaxy	A galaxy with no spiral arms — a huge, rounded swarm of stars.

Term	Definition
Event Horizon Telescope	A worldwide array of radio telescopes working together as one Earth-sized instrument.
Angular resolution	The smallest angle between two points that an instrument can tell apart.
Sextant	A navigation instrument for measuring the angle of a star or the Sun above the horizon.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
~2 years	Interval between Mars's conjunctions with the Sun
72,000 ft	Height of Olympus Mons — about 2.5 times Mount Everest
< 25 h	Length of a Martian day (sol)
< 1%	Mars's surface air pressure compared with Earth's at sea level
–225 to +70 °F	Range of Mars surface temperatures
93 bar	Venus's surface pressure — about 92 times Earth's
867 °F	Venus's surface temperature, day and night
~117 days	One Venusian day
~225 days	One Venusian year
16%	Weight reduction on Venus (10% gravity + 6% buoyancy)
4.51 / 4.54 billion yr	Age of the Moon / age of Earth
~15,000 mi	Earth-Moon distance when the Moon formed
~239,000 mi	Earth-Moon distance today, center to center
1.5 in/yr	Rate at which the Moon is moving away
½°	Angular diameter of the Moon (a pinky at arm's length is ~1°)
27.3 days	Sidereal month (and the Moon's rotation period)
29.5 days	Lunation, or synodic month — New Moon to New Moon
50-50	Chance of a Milky Way-Andromeda collision in the next 10 billion years
20 μas	Angular resolution of the Event Horizon Telescope
2019 / 2022	First images of the M87 and Milky Way black holes
~42°	Latitude of Chicago, and so the altitude of Polaris seen from there

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Mars, the Red Planet

Mars is something of a wanderer. About every two years it passes behind the Sun at **conjunction** and disappears for a while, then works its way back into the morning and evening sky. Its red color comes from **iron oxide** — rust — on its surface, and it is usually the fourth brightest object in our sky that shines by reflected sunlight, after the Moon, Venus, and Jupiter. Color plus brightness make it easy to spot.

Its two small moons, **Phobos** and **Deimos**, are named for the Greek gods of fear and dread. They may be captured asteroids, or they may have formed from debris after a giant impact on Mars; scientists are still debating. More important for any possible life, Mars has no active **magnetosphere**, so its surface is exposed to the **solar wind**. Leftover patches of magnetism survive in the southern highlands, and auroras occur all over the planet.

Mars's volcano **Olympus Mons**, now dormant, stands about 72,000 feet tall — roughly two and a half times the height of Mount Everest — and is the largest planetary volcano in the Solar System.

Mars's axial tilt and rotation are close to Earth's. A Martian day, a **sol**, is just under 25 hours, and Mars has four seasons. But at about one and a half times Earth's distance from the Sun, its year lasts a little under two Earth years, so each season is about twice as long as ours. The atmosphere is less than one percent as dense as Earth's at sea level and holds very little heat; surface temperatures range from about -225°F to $+70^{\circ}\text{F}$.

Part 2 — Venus, Earth's “Evil Twin”

Venus is the second planet from the Sun and, like Earth and Mars, a **terrestrial**, or rocky, planet. After the Moon it is the brightest object in our sky that shines by reflected sunlight. It orbits at about 0.7 **astronomical units** (one AU is the average Earth-Sun distance, about 93 million miles). Because it is closer to the Sun than we are — an **inferior planet** — it never strays far from the Sun in our sky, and we see it either as an evening star after sunset or as a morning star before dawn.

The ancients gave the two appearances different names. The evening star in the west was Hesperus to the Greeks and Vesper to the Romans; the morning star in the east was Phosphorus to the Greeks and Lucifer to the Romans — Latin for “light-bearer.”

Named for the goddess of love, Venus is anything but welcoming. Its atmosphere is 96 percent carbon dioxide and 4 percent nitrogen, at a crushing 93 **bar**, about 92 times Earth's sea-level pressure. A **runaway greenhouse effect** holds the surface at a steady 867°F , day and night — hot enough to melt lead and desolder a spacecraft's electronics. The clouds that cover the planet are droplets of sulfuric acid, in air laced with sulfur dioxide, and may produce lightning, though that is still debated.

Venus has no magnetic field to shield against the solar wind, though the solar wind does induce some auroral activity. Winds at the cloud tops reach about 220 miles per hour, but at the

surface they barely reach 2.5 miles per hour. The surface is dim, lit by only 5,000 to 14,000 **lux** compared with 100,000 lux on a bright Earth day.

Venus spins slowly and backward — **retrograde rotation**, clockwise as seen from above the Solar System. As a result one Venusian day, sunrise to sunrise, lasts about 117 Earth days, while its year is about 225 Earth days: only about two Venusian days per Venusian year.

One consolation for a visitor: you would weigh about 16 percent less. About 10 percent comes from Venus's slightly weaker gravity and about 6 percent from the **buoyant force** of its dense atmosphere. On Earth, air buoyancy is only an eighth of one percent of your weight — about a fifth of a pound for a 180-pound person. The Soviet **Venera** landers reached the surface, but the longest-lived lasted only about two hours before heat and pressure destroyed it.

Part 3 — The Moon's Origin and Retreat

The Moon is almost as old as Earth: about 4.51 billion years, compared with Earth's 4.54 billion. Most astronomers now agree it formed when a Mars-sized body called **Theia** struck the early Earth — not head-on but a glancing blow. Theia was vaporized along with a chunk of Earth, the debris formed a ring-like disk around Earth, and gravity pulled it together into the Moon. That explains why the Moon's makeup resembles Earth's mantle and why it lacks a large iron core.

The young Moon was much closer, probably about 15,000 miles from Earth, center to center. Through tidal action it has taken rotational energy from Earth, slowing Earth's spin and moving outward. Today the average distance is about 239,000 miles, roughly 16 times the original, and it is still growing by about 1.5 inches per year. We measure it precisely by bouncing lasers off reflectors left on the Moon by Apollo astronauts and Soviet rovers.

Sky observers describe size by angle. A pinky finger at arm's length covers about 1°; the Moon covers only about half that. However large the Moon looks — thanks to the Moon illusion — your pinky at arm's length easily covers it.

Part 4 — Tidal Locking, Months, and Phases

Billions of years of mutual tidal forces have left the Moon **tidally locked**: it shows only its near side to Earth. The far side is not permanently dark; we simply never see it from here. Locking means the Moon's rotation period and its orbital period are equal — both about 27.3 days measured against the distant stars, the **sidereal month**.

The month of phases is longer. A **lunation**, or synodic month, runs from one New Moon to the next. While the Moon circles Earth, Earth is also moving along its orbit, so after one trip around the Moon needs about two more days to line up with the Sun again. A lunation is 29.5 days.

The main phases are New Moon, First Quarter, Full Moon, and Third Quarter, with waxing crescent, waxing gibbous, waning gibbous, and waning crescent in between. Each shows up at predictable times: a First Quarter Moon is high in the south around sunset; a Full Moon rises in the east around sunset and stays up all night; a Third Quarter Moon rises around midnight and is high in the south around sunrise.

In the Northern Hemisphere the First Quarter Moon at sunset even hints at the season. Around the equinoxes its **terminator**, the shadow line, stands straight up and down. Near the Summer

Solstice it tilts back, as if the Moon were “smiling”; near the Winter Solstice it tilts forward, as if “frowning.”

Part 5 — The Milky Way and the Local Group

Our Solar System is an ordinary star-planet system in the **Milky Way**, a **barred spiral galaxy** — a “star city” — much like Messier 83, the Southern Pinwheel Galaxy. Until the 1920s many astronomers thought the Milky Way plus a few smudgy **nebulae** was the whole Universe. Edwin Hubble and others showed that many of those nebulae, especially the spiral ones, were separate, massive, distant galaxies, enlarging our picture of the Universe by many orders of magnitude.

The Milky Way belongs to the **Local Group**: more than 50 gravitationally connected galaxies, with more still being confirmed. The **Large** and **Small Magellanic Clouds** are small, irregular members easily seen from the Southern Hemisphere, named for Ferdinand Magellan, whose crew described them on his voyage around the world in the 1520s. The Milky Way and the Large Magellanic Cloud are expected to merge in about 2 billion years.

The largest member is the **Andromeda Galaxy**, M31, a spiral like the Milky Way but larger. A collision between the two was long expected; the latest studies put it at about a 50-50 chance over the next 10 billion years, and it would be more a gravitational dance reshaping both galaxies than a crash.

Part 6 — Beyond the Local Group: M87 and the Event Horizon Telescope

Farther out are different kinds of galaxies, such as the supergiant **elliptical galaxy** Virgo A, or **M87**. An elliptical galaxy has no spiral arms; it is a huge blob of stars held together by gravity. M87 hosts a **supermassive black hole** at its center.

In 2019 that black hole became the first ever imaged, by the **Event Horizon Telescope** (EHT), eight radio telescopes around the world working together. Its **angular resolution** of 20 micro-arcseconds is sharp enough to read a newspaper in New York from a sidewalk café in Paris. In 2022 the EHT also imaged the black hole at the center of our own Milky Way.

Part 7 — Finding Your Way by the Stars

Polaris, the North Star, sits almost directly over Earth's North Pole, so its angle above the northern horizon very nearly equals the observer's latitude — about 42° in Chicago.

Sailors used this on every clear night at sea, first with the **astrolabe** and later with the **sextant**, essentially a glorified protractor with mirrors and an eyepiece. When magnetic compasses first came into use, sailors did not understand how they worked, and many guessed the floating magnet was attracted to the North Star itself — people making sense of new technology with the sky knowledge they already had.

STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Mars, the Red Planet

1. What is a conjunction, and why does Mars periodically “go away” from our sky?

2. What gives Mars its red color, and which three sunlight-reflecting objects usually outshine it?

3. Name Mars's two moons and give the two proposed explanations for their origin.

4. Why does the lack of an active magnetosphere matter for Mars's surface?

5. Mars's day is close to Earth's, yet its seasons are about twice as long. Explain.

6. Why can Mars swing from about -225°F to $+70^{\circ}\text{F}$?

Part 2 — Venus, Earth's “Evil Twin”

1. Why is Venus seen only as an evening star or a morning star, never high in the midnight sky?

2. Give the Greek and Roman names for Venus as the evening star and as the morning star.

3. Describe Venus's atmosphere and surface conditions, and explain why the temperature is the same day and night.

4. Why is a Venusian day about 117 Earth days long, and how many of them fit in a Venusian year?

5. A visitor to Venus would weigh about 16% less than on Earth. Where do the two parts of that reduction come from?

6. What happened to the Soviet Venera landers, and why?

Part 3 — The Moon's Origin and Retreat

1. Describe the giant-impact model for the Moon's formation.

2. What two properties of the Moon does the giant-impact model explain?

3. Why has the Moon moved away from Earth, and what has happened to Earth as a result?

4. How fast is the Moon receding now, and how do we measure its distance so precisely?

5. How large does the Moon appear in degrees, and how can you check it with your hand?

Part 4 — Tidal Locking, Months, and Phases

1. What does it mean that the Moon is tidally locked? Is its far side always dark?

2. Distinguish a sidereal month from a lunation, and explain why the lunation is longer.

3. Name the four main and four minor phases of the Moon, in order starting from New Moon.

4. Where and when would you look for a First Quarter Moon, a Full Moon, and a Third Quarter Moon?

5. How can the terminator on a First Quarter Moon at sunset indicate the season in the Northern Hemisphere?

Part 5 — The Milky Way and the Local Group

1. What kind of galaxy is the Milky Way, and what is the Solar System's place in it?

2. What did Edwin Hubble and others discover in the 1920s, and why was it so important?

3. What is the Local Group? Describe the Magellanic Clouds and their future with the Milky Way.

4. What is the current outlook for a collision between the Milky Way and Andromeda?

Part 6 — Beyond the Local Group: M87 and the Event Horizon Telescope

1. How does an elliptical galaxy such as M87 differ from a spiral galaxy?

2. What did the Event Horizon Telescope achieve in 2019 and 2022, and how does it work?

3. What does an angular resolution of 20 micro-arcseconds mean in everyday terms?

Part 7 — Finding Your Way by the Stars

1. Why does the altitude of Polaris above the northern horizon give your latitude?

2. What instruments did sailors use to measure Polaris's altitude, and what mistaken idea did early sailors have about the compass?

GO OUTSIDE AND CHECK

Much of this episode can be checked by eye. Find Mars and Venus when they are visible, and note which horizon Venus is near. Cover the Moon with your pinky at arm's length. Watch the Moon's phase and rising time change over a week. And on a clear night, estimate the height of Polaris above the northern horizon with your fist at arm's length (about 10° per fist) — it should match your latitude.

YOUR NOTES

Notes